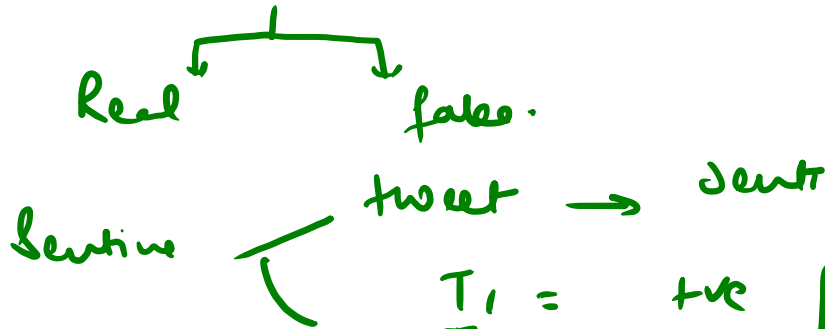


SNA

Deepfake Image Detection:

labelled - Input - output



False review detection

$R_1 \rightarrow$ false.
 $R_2 \rightarrow$ false.

Case 1:- Supervised - labelled data.

$\left| \begin{array}{c} \text{rec.} \\ \text{dev.} \\ \text{svm} \end{array} \right| \rightarrow$ Review.

⇒ unsupervised = unsupervised. uns.

Labelled data — trained model.

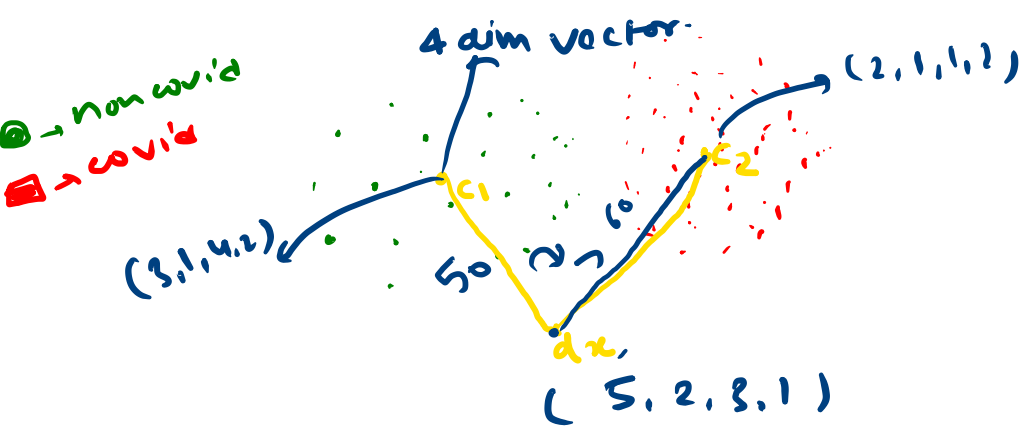
1000 $\left\{ \begin{array}{l} 100 \text{ test sample.} \\ 900 \text{ train} \end{array} \right.$

$\left\{ \begin{array}{l} S_0 \rightarrow C_1 \rightarrow 40 \text{ correct} \\ S_0 \rightarrow C_2 \rightarrow 40 \rightarrow \text{corr.} \end{array} \right.$
= 80 sample corr
20 sample incor

Unlabelled — clustering.

1000 $\left\{ \begin{array}{l} 200 \text{ cluster 1} \\ 300 \text{ cluster 2} \end{array} \right.$ = 80%.

Algorithm
 $\left\{ \begin{array}{l} \boxed{86} \text{ } \mu_1 \\ S_0 \text{ } S_1 \text{ } S_4 \text{ } S_5 \end{array} \right.$



$t_1 \ t_2 \ t_3 \ t_4$

$d_1 \rightarrow 4, 6, 8, 10$

$d_2 \rightarrow 3, 5, 9, 12$

$c_1 \rightarrow 102, 103$

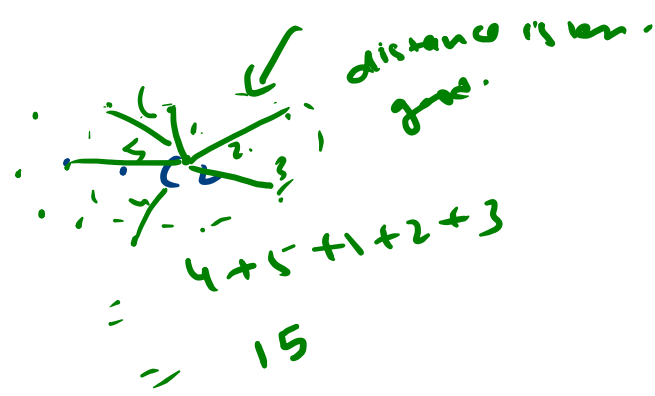
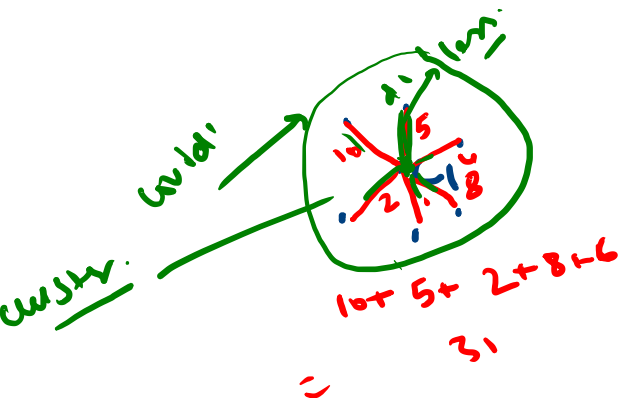
$101 \rightarrow 1$

Euc -

$$(c_1 \rightarrow dx) \quad \sqrt{(x_1 - x_1)^2 + (x_3 - x_4)^2 + (x_5 - x_6)^2}$$

$$= \sqrt{(5-3)^2 + (2-1)^2 + (3-4)^2 + (4-2)^2} = 50$$

$$(c_2 - dx) = \sqrt{(5-2)^2 + (2-1)^2 + (3-1)^2 + (4-2)^2} = 60$$



Intra cluster distance of
cluster 1

$$31 + 15 = 46$$

Intra cluster distance of
cluster 2

$$= \sum_{j=1}^K w_i \sum_{i=1}^N (C_k - d_i)$$

$w_i = 1$ if $d_i \in C_k$
 $w_i = 0$ if $d_i \notin C_k$

pos. neg. v.
 Jan
 class.
 Classification
 non der. su.
 numerical.

of +ve words x
 # of -ve words y
 # of words z

tweet length
 +ve, -ve, neu

(d1, c1)
 (d2, c1)
 (d1, c1)
 (d1, c1)

pos. neg. v.
 noun, no, adj. -

pos. terpsin.
noun, no. adjekt —

tweet length
+ve, -ve, neu

Social network.

node

GRAPH:-

Comments

Don

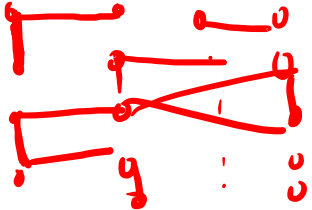
user

a

l_1
 l_2

edge.

edge.



venture

2 250

possible
Alors in



Any pair

1.2

3.4

S, G

Edge Betweenness

